



## RCM & Billing Decision Guide

Companion to the *Billing* docs.

### Section 1: Use Case & Participants

#### 1.1 What is your reimbursement model?

- Insurance / claims-based – you bill payers and reconcile their responses
- Patient self-pay / cash – patients pay you directly; no payer claim
- Value-based / capitation – per-member-per-month or risk-based
- Some combination

*Why: Claims-based reads every lane; self-pay skips the Charge → Claim → Payer lane; value-based leans on Foundations and the Value-Based lane (3.11).*

#### 1.2 How much of the revenue cycle runs in Medplum vs. delegated?

- **Clearinghouse-direct** – Medplum runs the cycle; the clearinghouse is transport. You own scrubbing, denials, and remittance posting. The most complete documented path.
- **RCM-automation partner** – you send encounter and charge data and delegate the back end; scrubbing, payer routing, denials, resubmission, patient collections, and reporting run on the partner's platform. The integration surface is claim submission; the partner's billing tasks sync back into Medplum as Tasks.
- **Documents only (no electronic claims)** – you don't run claims through a clearinghouse or RCM partner. Generate a CMS-1500 PDF for the

occasional paper/portal payer, or hand patients a superbill to self-submit for out-of-network reimbursement. Common for cash-pay and OON practices (see 1.1); also a per-payer fallback you can layer onto any engine above.

- **External Practice Management (PM) / billing system** – Medplum is the clinical source of truth; charges or claims hand off to an existing PM/billing system that owns the rest.

*In one line: a clearinghouse moves your claims; an RCM platform runs your revenue cycle. Either way Medplum owns the front half (coverage, charge capture, claim assembly); the paths diverge on the back half.*

| Revenue-cycle stage                 | Clearinghouse                | RCM partner                   | Documents only      | External PM |
|-------------------------------------|------------------------------|-------------------------------|---------------------|-------------|
| Coverage & charge capture (3.1–3.5) | You build                    | You build                     | You build           | You build   |
| Eligibility (3.3)                   | You build                    | Partner platform              | Manual              | External    |
| Claim assembly (3.6)                | You build                    | You build                     | You build           | External    |
| Scrubbing / validation              | You build                    | Partner owns                  | Light               | External    |
| Submission (3.6)                    | You build (837P)             | Partner submit op             | PDF / patient files | External    |
| Responses & ERA posting (3.7)       | You build (webhook + poller) | Partner owns; Tasks sync back | Manual              | External    |
| Denials & resubmission (3.8)        | You build                    | Partner owns; Tasks sync back | Manual              | External    |

| Revenue-cycle stage   | Clearinghouse | RCM partner  | Documents only | External PM |
|-----------------------|---------------|--------------|----------------|-------------|
| AR & reporting (3.10) | You build     | Partner owns | Manual         | External    |

### How to choose:

|           | Clearinghouse                                       | RCM partner                                                                   |
|-----------|-----------------------------------------------------|-------------------------------------------------------------------------------|
| You get   | Full control; billing data stays as FHIR in Medplum | A payer-rules engine and in-house billers; far less to build                  |
| Trade-off | You build and maintain the entire back office       | Recurring fees; less control; AR and reporting live on the partner's platform |
| Choose if | You have RCM expertise and want a fully owned cycle | You want to outsource the cycle                                               |

*Paths can mix. The choice sets which Section 3 lanes you build (3.5–3.10) vs. read only for the boundary.*

### 1.3 Who does the billing work, and is the patient part of it?

- Charge capture: clinicians at point of care, or back-office entry?
- Coding and claim work: dedicated coders/billers, or automated with exception review?
- Accounts receivable (AR) follow-up: who works denials, aging, and patient balances?
- Patient: do patients see estimates, statements, or pay online?

*Why: drives charge-capture design (3.5), who works denials and AR (3.8, 3.10), and whether patient-facing billing is in scope (3.9).*

### 1.4 Where does coding happen, and how automated is it?

- Auto-derived from clinical data (visit type, orders, procedures)?
- Coder-reviewed before the claim is built?
- Fully manual?

*Why: drives the coding and review model in charge capture (3.5) and where scrubbing exceptions surface (3.6).*

### **1.5 New build or replacing/extending, and what else must billing touch?**

- If replacing: what billing/PM system, and what gaps drove the change?
- If new: is billing core to the product (you're building an RCM/billing product) or additive (a clinical app that also needs to bill)?
- What other systems must billing read from or write to (clinical/EHR source, PM, clearinghouse, accounting/GL, payment processor, payer portals)?

*Why: surfaces integration boundaries and the source-of-truth split behind 1.2, 3.6, and 3.7. Core-to-product builds tend to own the cycle (a clearinghouse); additive builds tend to delegate (an RCM partner or external PM).*

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## **Section 2: Feature Scoping**

*Go through each row together. For each cell under Yes / No / Nice-to-have / Not sure, mark the customer's answer. The § column points to the deep-dive subsection that covers each feature.*

| # | Feature                                     | §   | Yes | No | Nice-to-have | Not sure |
|---|---------------------------------------------|-----|-----|----|--------------|----------|
| 1 | Coverage & payer modeling                   | 3.1 |     |    |              |          |
| 2 | Patient financial account / running balance | 3.2 |     |    |              |          |
| 3 | Real-time eligibility & benefits (270/271)  | 3.3 |     |    |              |          |
| 4 | Prior authorization (pre-service)           | 3.4 |     |    |              |          |

| #  | Feature                                                     | §           | Yes | No | Nice-to-have | Not sure |
|----|-------------------------------------------------------------|-------------|-----|----|--------------|----------|
| 5  | Patient cost / Good-Faith Estimates                         | 3.4         |     |    |              |          |
| 6  | Charge capture (encounter-linked) & pricing / fee schedules | 3.5         |     |    |              |          |
| 7  | Coding (CPT/ICD-10, modifiers) with review queue            | 3.5,<br>3.6 |     |    |              |          |
| 8  | Assemble, validate & submit claims                          | 3.6         |     |    |              |          |
| 9  | CMS-1500 PDFs / superbills (manual / patient submission)    | 3.6         |     |    |              |          |
| 10 | Ingest payer responses & post remittance (277CA / 835)      | 3.7         |     |    |              |          |
| 11 | Denials, corrected claims & appeals                         | 3.8         |     |    |              |          |
| 12 | Patient statements, invoices & online payment               | 3.9         |     |    |              |          |
| 13 | Claim / AR status, aging & reporting                        | 3.10        |     |    |              |          |
| 14 | Value-based contracts (panel, quality measures, capitation) | 3.11        |     |    |              |          |

## Section 3: Feature Deep Dives

Cover each feature flagged Yes or Nice-to-have in Section 2. The goal is to land on a clear recommended approach by the end of each section.

Section 3 is grouped into four lanes:

| Lane                                | Subsections | When to read                                         |
|-------------------------------------|-------------|------------------------------------------------------|
| <b>Foundations</b>                  | 3.1 – 3.2   | Always                                               |
| <b>Pre-service / Patient Access</b> | 3.3 – 3.4   | When you verify coverage or authorize before service |
| <b>Charge → Claim → Payer</b>       | 3.5 – 3.8   | When you bill insurance                              |
| <b>Patient &amp; Operations</b>     | 3.9 – 3.10  | Always                                               |
| <b>Value-Based</b>                  | 3.11        | When you have capitated or risk-based contracts      |

### Foundations

#### 3.1 Coverage & Insurance Model

How you represent a patient's insurance (the coverage stack, the subscriber, payers, cost-sharing) is the foundation every claim and eligibility check builds on. See [Representing Patient Insurance Coverage](#); coverage capture at intake is in the Intake guide (3.5).

#### Questions:

- Do patients commonly have more than one coverage, and must you bill them in order?
- Is the subscriber sometimes someone other than the patient?
- Is your payer list curated in Medplum, synced, or created on demand?
- What cost-sharing (copay, coinsurance, deductible) do you need for estimates and patient billing?

| Situation                 | Approach                                                                                                                                                                                                                                                                   |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subscriber is the patient | One Coverage; subscriber = Patient; relationship = self; payor → Organization.                                                                                                                                                                                             |
| Primary + secondary       | One Coverage per plan; Coverage.order sets billing sequence. Secondary submits after the primary remittance posts (3.7).                                                                                                                                                   |
| Subscriber ≠ patient      | Coverage + RelatedPerson for the subscriber. The relationship inverts: Coverage.relationship is the patient's relationship to the subscriber, the RelatedPerson's is the reverse (coverage child → related person parent). Getting this backwards is a common billing bug. |
| Member ID & plan details  | Coverage.subscriberId for member ID; Coverage.class for group, plan name, RxBIN/RxPCN.                                                                                                                                                                                     |
| Cost-sharing              | Coverage.costToBeneficiary; feeds estimates (3.4) and patient responsibility (3.9).                                                                                                                                                                                        |
| Payer directory           | Medplum ships a built-in <b>Payer Directory</b> of Organizations, or curate your own. The payer identifier system differs by backend; see 3.6.                                                                                                                             |
| Card images               | Store as DocumentReference/Binary linked to the Coverage.                                                                                                                                                                                                                  |

### 3.2 Account & Financial Ledger

Decide whether you track a patient financial account that rolls charges, claims, and payments into a balance (the backbone for AR in 3.10 and patient billing in 3.9), or let each charge stand alone.

*Note: the rows below are one FHIR approach to a financial ledger; validate against your reporting and reconciliation requirements.*

### Questions:

- Do you need a single balance rolling up charges, payer payments, adjustments, and patient payments?
- Is the account scoped per patient, per visit, or per episode?
- Do you track payer and patient responsibility separately on the same account?
- Must the account reconcile against an external accounting/GL system?

| Situation         | Approach                                                                                                                |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|
| Running balance   | Account as the ledger anchor; ChargeItem, Claim, PaymentReconciliation, and Invoice reference it.                       |
| Account scope     | Account.subject → Patient, or tie to Encounter for visit-level; apply one scope consistently.                           |
| Payer vs. patient | Payer side via claim/remittance (3.7), patient side via Invoice (3.9), both rolling to the same Account.                |
| No ledger         | Skip Account for pure pass-through submission where another system owns the balance, but you lose in-Medplum AR (3.10). |

## Pre-service / Patient Access

### 3.3 Eligibility & Benefits

Decide whether you verify coverage before service, what you ask the payer, and when the check runs. See [Insurance Eligibility Checks](#).

*Note: Eligibility is provided by the clearinghouse integration. An RCM partner may offer its own eligibility on its platform, outside that integration.*

### Questions:

- What must you confirm: active coverage, covered service, in-network status, or benefit/cost detail?
- When does the check run: scheduling, check-in, or batch pre-visit?
- Do you re-check on a cadence for ongoing patients?

- Who works failed or ambiguous checks?

| Situation                  | Approach                                                                                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verify coverage / benefits | CoverageEligibilityRequest (purpose: benefits); invoke the clearinghouse eligibility operation to run the X12 270/271 and return a CoverageEligibilityResponse. |
| Scope limit                | The current integration returns general benefits only, not arbitrary service-specific benefit queries.                                                          |
| In-network check           | Read network participation from the response (in-network indicator Y/N/U/W); run at provider selection or as a pre-schedule gate.                               |
| Cost-sharing               | Copay, coinsurance, remaining deductible, and out-of-pocket land in CoverageEligibilityResponse.insurance.item; the input to patient estimates (3.4).           |
| When it runs               | A Bot issues the request on the triggering event; persist the response for the visit and billing.                                                               |
| Exception                  | On failure or ambiguity, create a Task for registration rather than proceeding silently.                                                                        |

### 3.4 Prior Authorization & Estimates

Decide whether services need payer authorization before they happen, how you track that to a decision, and whether you produce patient cost estimates up front.

*Note: the eligibility- and Claim-based rows below are one FHIR approach to prior auth; validate against your payer requirements. The standards-based CMS-0057-F path (CDS Hooks / CRD) is in [alpha](#). Prior auth in a referral context is in the Referrals guide (3.7).*

#### Questions:

- Do any services require prior authorization before scheduling or delivery?
- How do you check whether auth is required, and track a submitted auth to approval/denial?
- Must scheduling be gated until authorization clears?
- Do you owe patients a cost estimate before service (e.g. a Good-Faith Estimate)?

| Situation                    | Approach                                                                                                                                                         |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is auth required?            | CoverageEligibilityRequest with purpose: auth-requirements; the response indicates whether authorization is required.                                            |
| Submit a prior auth          | Claim with use: preauthorization (distinct from a claim for completed services); ClaimResponse carries the decision. A Bot converts FHIR ↔ X12 for transmission. |
| Standards-based (CMS-0057-F) | Electronic prior auth via CDS Hooks / Coverage Requirements Discovery (CRD); <b>alpha</b> . The emerging payer-interop path as the rule phases in.               |
| Gate scheduling              | Hold in an auth-pending Task.businessStatus until the ClaimResponse approves, then release.                                                                      |

**Good-Faith Estimates (No Surprises Act).** One FHIR approach: build the estimate from pricing rules (ChargeItemDefinition/\$apply, 3.5) and the patient's cost-sharing (Coverage.costToBeneficiary, 3.1), delivered as a DocumentReference; validate against your compliance requirements. Mainly for self-pay and out-of-network patients.

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## Charge → Claim → Payer

*Insurance-claims-specific. Self-pay builds skip to Patient & Operations; pricing (3.5) still applies to self-pay rates and estimates.*

### 3.5 Charge Capture, Coding & Pricing

Decide what gets billed, where the codes come from, and how each line is priced. Coding is the largest source of denials, so treat it as first-class, not a byproduct of charge entry.

#### Questions:

- What generates a charge: a completed encounter, an order/procedure, a fulfilled service?
- Where do codes come from (CPT/HCPCS, ICD-10, modifiers): auto-derived, coder-reviewed, or manual (1.4)?
- Is pricing a single fee schedule, or does it vary by payer contract?
- Who reviews coding before a claim is built, and how do exceptions surface?

| Situation            | Approach                                                                                                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What gets billed     | One ChargeItem per billable line; ChargeItem.code carries CPT/HCPCS. Link context → Encounter and reference the Account (3.2) at creation, or reconciling into claims and AR later gets painful. |
| Pricing              | ChargeItemDefinition holds the rules; \$apply calculates the price.                                                                                                                              |
| Payer-specific rates | Multiple ChargeItemDefinitions keyed to payer/contract; apply the one matching the coverage.                                                                                                     |
| Coding source        | Auto-derive via a Bot, or set by a coder; route low-confidence items to a review Task.                                                                                                           |

| Situation | Approach                                                                                                           |
|-----------|--------------------------------------------------------------------------------------------------------------------|
| Diagnoses | Capture ICD-10 diagnoses to attach to the claim ( <code>Claim.diagnosis</code> , with item-level pointers) in 3.6. |

### 3.6 Claim Assembly & Submission

Decide how you build a valid claim from charges and coverage, and how it leaves your system. The 1.2 engine decision plays out here.

**⚠ One-way door:** Choosing a submission backend is hard to reverse. Each maps the `Claim` to a different format (837P, the partner's API, a CMS-1500 PDF) with a different response pipeline (3.7), and payer identifier systems differ. Keep one `Claim` as the source of truth so the backend stays swappable.

#### Questions:

- Are you billing as an organization or an individual provider?
- Which submission path(s), and does it vary by payer (electronic for most, paper/portal for some)?
- What validation must pass before submission, and where do failures surface?
- Do you submit secondary claims after the primary pays (coordination of benefits)?

| Situation        | Approach                                                                                                                                                                                                   |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Build the claim  | <code>Claim</code> referencing the Coverage ( <code>Claim.insurance</code> ), charges ( <code>Claim.item</code> ), and ICD-10 diagnoses ( <code>Claim.diagnosis</code> ) with item-level pointers.         |
| Billing provider | Set on <code>Claim.provider</code> , but backends read it differently (billing provider vs. rendering provider with the billing Organization via <code>PractitionerRole</code> ); confirm per integration. |

| Situation                          | Approach                                                                                                                                                                                            |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electronic (you own the cycle)     | The clearinghouse submit operation sends an 837P and writes a correlation id back to the Claim.                                                                                                     |
| RCM partner                        | The partner's submit operation returns a ClaimResponse; the partner owns scrubbing, denials, resubmission, and remittance on its platform, and billing tasks sync back into Medplum as Tasks (3.8). |
| Manual / paper / portal            | Claim/\$export generates a CMS-1500 PDF (Binary; experimental); log the outbound document.                                                                                                          |
| Patient-submitted (out-of-network) | Generate a superbill for the patient to file.                                                                                                                                                       |
| Validation                         | Validate before submit; surface a Task and don't transmit invalid claims. (In the current integration, a submission failure marks the Claim status = error.)                                        |
| Multiple channels                  | One Claim is the source of truth; run separate send paths per payer without diverging content.                                                                                                      |

### 3.7 Payer Responses & Remittance Posting

Decide how you ingest what the payer sends back (the 277CA acknowledgment, accepted or rejected pre-adjudication, and the 835 ERA remittance, what was paid, adjusted, or denied line by line) and how payments post. These share one pipeline, so design them together.

*Note: applies to the clearinghouse path, and the row specifics reflect the current integration. With an RCM partner, the partner owns response handling and posting; what flows back is its billing Tasks.*

**Questions:**

- How do responses arrive: webhook push, polling, or partner API?
- How do you match a response to the originating claim?
- How do payer payments and adjustments post to the account?
- What share posts automatically vs. needs review?

| Situation              | Approach                                                                                                                                                                                                            |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ingestion              | A webhook bot (primary) plus an optional poller (catch-up). Make it idempotent on the payer's inbound transaction id (not the webhook delivery id); webhooks redeliver, and without dedup you double-post payments. |
| Acknowledgment (277CA) | Persist as a ClaimResponse; surface front-end rejections to denials (3.8).                                                                                                                                          |
| Remittance (835 ERA)   | Create a PaymentReconciliation and ClaimResponse; the ERA PDF is stored as a DocumentReference. Post payments/adjustments against the Account (3.2).                                                                |
| Matching               | Match on the patient-control-number (PCN) assigned at submission from the Claim id (835), or the correlation id (277).                                                                                              |
| Claim status           | Status lives on ClaimResponse; query ClaimResponse?request=Claim/{id} (3.10).                                                                                                                                       |
| Review                 | Auto-post clean remittances; route mismatches to a Task.                                                                                                                                                            |

### 3.8 Denials, Rejections & Resubmission

Decide how you work claims the payer rejects or denies: correcting and resubmitting, or appealing.

*Note: The rows below are one FHIR approach to denial handling; validate against your workflow. On the clearinghouse path you build this in Medplum; with an RCM partner, denials are worked on the partner platform and surface back as synced Tasks.*

#### Questions:

- Do you work front-end rejections (277CA) differently from adjudicated denials (835)?
- When you fix and resend, is it a corrected claim or a fresh submission?
- Do you track appeals (deadlines, supporting documentation, outcome)?
- How do staff find claims that need work?

| Situation       | Approach                                                                                               |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Surface denials | Denial detail lives on the ClaimResponse (3.7); create a Task work queue for the biller.               |
| Corrected claim | Resubmit a corrected Claim linked to the original (Claim.related), via the same backend (3.6).         |
| Appeals         | Track as a Task with deadlines; attach supporting docs as DocumentReference.                           |
| Find work       | Search over open denial Tasks (3.10).                                                                  |
| Partner-owned   | The RCM partner works denials/resubmission on its platform; surfaced tasks sync into Medplum as Tasks. |

## Patient & Operations

### 3.9 Patient Responsibility & Self-Pay

Decide how you bill patients for their share (copays, coinsurance, deductibles, full self-pay) and how they pay.

#### Questions:

- What balances do you bill: post-adjudication responsibility, self-pay, or both?
- Do you issue statements, and on what cadence?
- Do you take online payment / card on file, through which processor?
- For out-of-network patients, do you provide a superbill instead of billing the payer?

| Situation       | Approach                                                                                                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------|
| Patient balance | Invoice for patient-owed amounts rolling to the Account (3.2); responsibility from remittance (3.7) or self-pay pricing (3.5). |
| Online payment  | Integrate a payment processor; a <a href="#">sample integration</a> shows keeping invoices and payments in sync.               |
| Self-pay / cash | Price via ChargeItemDefinition/\$apply (3.5); invoice the patient directly, no payer claim.                                    |
| Out-of-network  | Provide a superbill (3.6) for the patient to submit for reimbursement.                                                         |

### 3.10 Accounts Receivable, Aging & Lifecycle Status

Decide how you track where every claim and balance sits, and how staff find work that's stuck or overdue.

*Note: The rows below are one FHIR approach to AR tracking; validate against your reporting needs. Most relevant on the clearinghouse path; an RCM partner owns AR and reporting.*

#### Questions:

- What lifecycle states matter (e.g. draft → submitted → accepted → paid → denied → closed)?
- How do you standardize states so queues, reports, and integrations agree?
- Do you track aging (days since submission, days pending) and AR buckets?
- How do staff find overdue or stuck claims and balances?

| Situation        | Approach                                                                                                                                                |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lifecycle states | Track progress with <code>Task.businessStatus</code> ; move <code>Task.status</code> on clear transitions. Keep the state set small and single-sourced. |
| External signals | Bots/subscriptions advance status when responses arrive (3.7): accepted, paid, denied.                                                                  |
| Aging & AR       | Search over <code>Claim.created</code> , <code>Task.lastModified</code> , and Account balances; bucket by age.                                          |
| Overdue work     | Scheduled Bots or subscriptions flag items past an SLA into work queues.                                                                                |

## Value-Based

### 3.11 Value-Based & Capitation

Value-based care isn't a claims pipeline; it splits into measurement (quality, gaps in care) and money (capitation, risk, settlement). Read only for capitated or risk-based contracts.

*Note: Quality measurement uses documented operations; the financial rows are one FHIR approach to validate against your contracts. Quality/HEDIS reporting overlaps clinical analytics; see [Analytics](#).*

#### Questions:

- Which contracts are capitated or risk-based, and which members are attributed to you?

- Which quality measures (eCQM, HEDIS, stars) are you accountable for?
- Do you still submit claims or encounter data when capitated (for risk adjustment and measurement)?
- How are capitation payments and any shared-savings settlement reconciled?

| Situation                          | Approach                                                                                                    |
|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Attribution / panel                | Group as the member roster; ingest the payer roster via a Bot; CareTeam for accountable teams.              |
| Quality measurement & gaps in care | Measure + MeasureReport via Measure/\$evaluate-measure; eCQM and HEDIS via <a href="#">Analytics</a> .      |
| Risk adjustment (HCC / RAF)        | Capture Condition coding from encounters and submit as encounter data; ties to coding (3.5).                |
| Encounter data                     | Even when capitated, reuse claim assembly (3.6) to send 837/encounter data for measurement and risk.        |
| Payments (PMPM, settlement)        | Model received capitation and shared-savings settlement as PaymentReconciliation against the Account (3.2). |

## Related Integrations

The billing and revenue-cycle partners Medplum integrates with first-party:

- **Stedi** — X12 [eligibility checks](#) (270/271) and [professional claim submission](#) (837P) with [claim responses](#).
- **Candid Health** — revenue cycle automation and professional claim submission from FHIR Claim resources.